I. Syllabus and Grading Structure (15-20 min)
   A. Introductions
      1. Instructor
         - Name: Luke Miller
         - Degree: MS CS
         - Title: GRA
         - Contact: ljmbm5@umsystem.edu
         - Office: FH504
         - Hours: By appointment
      2. Supervising
         - Name: Yugyung Lee
         - Degree: PhD
         - Title: Prof
         - Contact: leeyu@umkc.edu
         - Hours: Tue/Thu 4-5 PM
         - Office: FH560D
      3. Icebreaker
         - Backgrounds
         - Math familiarity
   B. Course Overview
      1. Title
         - CS490/5590
         - QC Apps in DS/AI/DL
         - 3 credits
         - In-person
      2. Description
         - QC foundations
         - Algos
         - Qiskit
         - DS/AI focus
         - QSVM
         - QNN
      3. Learning Outcomes
         a. Quantum circuits
            - Build
            - Manipulate
         b. Algos
            - Grover's
            - Shor's
         c. Optimization
            - QAOA
            - VQE
         d. QML
            - QSVM
            - QCNN
            - Feature selection
         e. Hybrid
            - QC-classical
            - ML pipelines
         f. Project
            - Quantum-enhanced
            - AI/DS
   C. Logistics and Policies
      1. Modality
         - In-person
         - Closures: Recordings/reschedule
      2. Tech
         1. Jupyter/Qiskit
            - Colab
            - Kaggle
            - No local support
         2. LaTeX
            - Overleaf
            - No local support
      3. Materials
         - Free online
         - No textbooks
      4. Late Work
         - None without emergency
         - Notify early
      5. Philosophy
         - Learning
         - Exploration
         - Effort over perfection
      6. Attendance Verify
         - Points for attendance
         - Necessary for success
   D. Evaluation and Grading

1. Total
   - 1000 pts
   - 0.1% each

   a. Project: 600 pts
      - Check-ins: 3x50=150
      - Hackathon/Fall Fest: 250
      - Submission/Presentation: 300

   b. HW/Quizzes: 220 pts
      - HW: 8x15=120
      - Quizzes: 5x20=100

   c. Attendance: 180 pts
      - 7 pts/session
2. HW
   - In-class tutorials
   - 2-3 hrs
   - Completion + discussion
   - Show HW0: Math/Qiskit
3. Quizzes
   - In-class start/review
   - 2 attempts
   - Password
   - No makeups
4. Exams
   - None
   - Hands-on focus
5. Scale
   - A: 94-100
   - A-: 90-93
   - ...
   - F: ¡60

E. Schedule Highlights

1. Start
   - 8/26
   - Today: Topic 0 - Math Review
2. Keys
   - HW0 due 9/2
   - Quiz 1 9/4
   - Hackathon 11/14
   - Presentations 12/9-12/11
3. Topics
   - QC intro: 8/28
   - Circuits: 9/2
   - Entanglement: 9/9
   - Algos: 9/11-9/30
   - QML: 10/7+
4. Notes
   - No class 11/13
   - Recorded 11/11 (Veterans Day)

F. Concerns

- Address briefly
- E.g., attendance: Rewards engagement
- Project: For application

II. Quantum Motivation (10-15 min)

A. Why QC?

1. Classical limits
   - Expo problems
   - Factoring
   - Simulations
2. Quantum edge
   - Poly time
   - Drugs
   - Optimization
3. AI tie
   - ML enhancement
   - Training
   - Kernels

B. Quantum Advantage

1. Principles
   - Superposition
   - Entanglement
   - Interference
2. Algos
   - Grover's: Search
   - Shor's: Factoring
   - QAOA/VQE: Opt
3. Example
   - Google Sycamore 2019

- 53 qubits
- 200s vs 10k yrs
- IBM 2023+: 100+ qubits

C. Tech Advancements

1. Hardware
   - 5 qubits (2016)
   - 1000+ planned
   - IBM Condor
2. Software
   - Qiskit
   - Hybrid clouds
   - AWS/Google
3. AI
   - QML: QGNNs, K-Means
   - Apps: Finance/pharma

D. Hook

- Project build
- HW0 Qiskit on IBM

E. Poll

- QC/Python experience?

III. Math Review: Linear Algebra and Probability for Quantum (45 min)

A. Foundations Importance (3-4 min)

1. LA
   - States: Vectors
   - Gates: Matrices
   - Multi: Tensors
2. Prob
   - Measurements
   - QML uncertainty
3. Course
   - Sessions 1-3: Qubits
   - 4-7: Algos
   - 9-21: QML
4. Resources
   - Matrix Cookbook: LA (use numpy)
   - Ross: Prob
   - Nielsen/Chuang: Ch.2
   - NumPy/Qiskit
5. Note
   - Handout: Session0Handout
   - HW0: Code inners

B. Vectors/Hilbert (8-10 min)

1. Vectors
   - In C^n
2. Qubit
   - $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
3. Normalization
   - $|\alpha|^2 + |\beta|^2 = 1$
4. Hilbert
   - Inner: $\langle u|v\rangle = u^\dagger v$
5. Norm
   - $\sqrt{\langle v|v\rangle}$
6. Basis
   - Orthonormal
   - $|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$
   - $|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$
7. AI
   - Feature encoding
8. Ex
   - $|v\rangle = \begin{bmatrix}1\\i\end{bmatrix}$
   - Norm: $\sqrt{1\cdot 1 + |i|^2 \cdot 1} = \sqrt{2}$
   - Norm'd: $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\i\end{bmatrix}$
9. Interactive
   - Practice Problem 1 (Handout)
   - $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

   a. Board
   - Hand compute

   b. Solution
   - Norm: $|\psi| = \sqrt{\langle\psi|\psi\rangle} = \sqrt{\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 1} = 1$
   - $\langle Z\rangle = \langle\psi|Z|\psi\rangle = \frac{1}{2}\begin{bmatrix}1 & -i\end{bmatrix}\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}1\\i\end{bmatrix} = 0$

- Implies: Equal chance of +1/-1 measurements

c. Note
- HW0 Part A
- inner()
- normalize()
- NumPy

C. Matrices (8-10 min)

1. Operators
   - $C^n \to C^n$
2. Hermitian
   - $A^\dagger = A$
3. Unitary
   - $U^\dagger U = I$
4. Paulis
   - $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
   - $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
   - $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
5. Evals
   - $A|v\rangle = \lambda|v\rangle$
6. Ops
   - Add
   - Mult
   - Transpose
7. AI
   - Gates
   - VQE Hamiltonians
8. Ex
   - X: Herm/unit (Check $X^\dagger = X$, $X^\dagger X = I$)
   - Z: Evals +1/-1
   - Evecs: $|0\rangle$/ $|1\rangle$
9. Interactive
   - Compute Z eig
   - NumPy: np.linalg.eig(Z)

   a. Group
   - 1 min try

   b. Solution
   - Evals: [1, -1] (sorted real)
   - Evecs: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (up to phase; projectors $|0\rangle\langle 0|$, $|1\rangle\langle 1|$)

   c. Note
   - HW0 Part B
   - is$_hermitian/unitaryOnPaulis/H$

D. Tensors/Dirac (8-10 min)

1. Tensors
   - Multi-qubits
   - $|\psi 1\rangle \otimes |\psi 2\rangle$
   - Dim 4
2. Ex
   - $|01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$
   - $X \otimes I = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$
3. Dirac
   - Ket: $|v\rangle$
   - Bra: $\langle v|$
   - Inner
   - Outer: $|u\rangle\langle v|$
4. Props
   - Outer dagger: $|v\rangle\langle u|$
   - Density: $\rho = \sum p_i |\psi_i\rangle\langle\psi_i|$
5. AI
   - QNN states
   - Kernels
6. Interactive
   - Practice Problem 2 (Handout)
   - $|\psi\rangle = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

   a. Board
   - Vector/matrix build

   b. Solution

- $|\psi\rangle = \begin{bmatrix}1\\0\end{bmatrix} \otimes \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\\0\end{bmatrix}$
- $X \otimes I = \begin{bmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{bmatrix}$
- Result: $\frac{1}{\sqrt{2}}\begin{bmatrix}0\\0\\1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$

c. Note
- HW0 Part C
- kron()
- CNOT
- Bell

7. Quick
   - Practice Problem 3 (Handout)
   - Compute $|0\rangle\langle 1|$
   - Apply to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

   a. Solution
   - $|0\rangle\langle 1| = \begin{bmatrix}1\\0\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix} = \begin{bmatrix}0&1\\0&0\end{bmatrix}$
   - Apply: $\begin{bmatrix}0&1\\0&0\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\end{bmatrix} = \frac{1}{\sqrt{2}}|0\rangle$
   - Physical: Projection operation

E. Probability (8-10 min)

1. Classical
   - p_i sum=1
   - $E[X] = \sum x_i p_i$
2. Quantum
   - Born: $P(|i\rangle) = |\langle i|\psi\rangle|^2$
   - Density: $P(i) = \langle i|\rho|i\rangle$
   - Expect: $\langle A\rangle = \mathrm{Tr}(A\rho)$
3. Measurements
   - Projectors: $P_i = |i\rangle\langle i|$
   - Variance: $\Delta A = \sqrt{\langle A^2\rangle - \langle A\rangle^2}$
4. AI
   - QML outcomes
   - QSVM
   - Bayes
5. Ex
   - Z on $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
   - $\langle Z\rangle = \langle\psi|\begin{bmatrix}1&0\\0&-1\end{bmatrix}|\psi\rangle = 0$
6. Interactive
   - Practice Problem 4 (Handout)
   - $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

   a. Group
   - Calc probs/expect

   b. Solution
   - Probs: $P(|0\rangle) = \left|\frac{\sqrt{3}}{2}\right|^2 = \frac{3}{4}$, $P(|1\rangle) = \left|\frac{1}{2}\right|^2 = \frac{1}{4}$
   - $\langle X\rangle = \langle\psi|\begin{bmatrix}0&1\\1&0\end{bmatrix}|\psi\rangle = \frac{\sqrt{3}}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2} \approx 0.866$

   c. Note
   - HW0 Parts D-E
   - Projectors/density
   - NumPy sampling

F. Wrap-Up: Q&A/Expectations (3-5 min)

1. Math
   - Builds to QML/projects
2. HW0
   - Exercises + Qiskit
   - Due 9/2
   - Show Bell
3. Q&A
   - Unfamiliar topics?